SMART WATER APP STUDENT GUIDE

GitHub | Clone | Open in VS Code | Modify with Copilot

Use this guide to get the Smart Water App running locally, explore its modules, and make a safe first change. The app uses Vite for development and keeps its data and building model in the repository.

1. Before you begin

- The GitHub repository is public. You do not need a GitHub account, invitation, or individual permission to view, clone, or download the code. GitHub sign-in may still be needed to use Copilot.
- Install Git: https://git-scm.com/downloads. On Windows, Git includes Git Bash.
- Install Visual Studio Code: https://code.visualstudio.com/.
- In VS Code, install the GitHub Copilot / Copilot Chat extension if available to you, and sign in when prompted. If you do not have Copilot access, ask your teacher about an alternative AI coding assistant.

Public repository: https://github.com/smart-water-EDU/SmartWaterApp

2. Clone the repository - choose ONE method

Method A: Inside VS Code (recommended)

Open the public SmartWaterApp repository on GitHub. Click Code, select HTTPS, and copy the repository URL.

2. Open VS Code. Press Ctrl+Shift+P and choose Git: Clone (or Clone Repository).

3. Paste the copied HTTPS URL. GitHub sign-in is not required to clone this public repository.

Choose a permanent location on your computer. When cloning finishes, select Open to open the SmartWaterApp folder.

Clone only once. Later, reopen the same folder in VS Code using File > Open Folder.

Method B: Using Git Bash (alternative)

Open Git Bash and run the commands below. No GitHub sign-in is required to clone the public repository.

```
cd ~/Documents
git clone https://github.com/smart-water-EDU/SmartWaterApp.git
cd SmartWaterApp
```

In VS Code, select File > Open Folder and open the newly cloned SmartWaterApp folder.

If Documents is in OneDrive or another location, use File Explorer to choose a suitable folder and open Git Bash there instead.

3. Install and run the application

1. Install Node.js 22.12 or newer from https://nodejs.org/ if it is not already installed. Restart VS Code after installation.

2. In VS Code, open Terminal > New Terminal. Confirm the terminal is in the SmartWaterApp project folder, then run: npm install

3. Start the app with: npm run dev. Vite opens the app in your browser at http://127.0.0.1:5173/. Keep the terminal running while you work.

4. Edit the source in VS Code. Vite refreshes the browser after most changes. To stop the app, focus the terminal and press Ctrl+C. You can create a production build with npm run build.

The application runs on your computer. Its simulated readings, building model, graphics and student guides are included in the repository. The development server is local; the app does not need a cloud account or external data service.

4. Make a safe first change with Copilot

1. In VS Code, open the Copilot Chat panel (or the Chat icon) and make sure it can read your project folder.

2. Start with the read-only prompt below to understand the code before changing it.

Choose one small styling experiment. Ask Copilot to explain the planned change first, then approve only CSS edits.

Review changed files in the Source Control panel. Save the CSS file and let Vite refresh the browser to show the change.

Prompt 1 - Understand the project (no edits)

"Read README.md and inspect the project structure. Briefly explain which files control the visual style and how I can preview changes in the browser. Do not modify any files."

Prompt 2 - Change one color (CSS only)

"Make a small visual-only change: slightly soften the background color of the dashboard cards toward blue-gray. Change CSS only (look in src/styles/). Do not edit HTML, JavaScript, data files, charts, layout dimensions, labels or application logic. Tell me which CSS selectors you plan to change before making edits."

Prompt 3 - Improve a hover state (CSS only)

"Identify the CSS rule for the main tab buttons. Make the hover color a little clearer without changing button size, position or behavior. Edit CSS only. Do not modify src/ application code or data/. Explain the exact change."

Prompt 4 - Undo or compare

"Show me how to inspect the current file changes in VS Code Source Control and how to revert only the styling changes I made. Do not run destructive commands or modify any files without my approval."

5. Keep your work safe

- The application code is organized under src/ and the editable building, profile and monitoring files are in data/. For your first experiment, change CSS only under src/styles/.
- Do not rename or delete index.html, README.md, or project folders. Do not ask Copilot to rebuild or redesign the whole application.
- Check VS Code Source Control or run git status to see the changed files; use git diff to inspect exactly what changed.
- If the change is not good, use Undo (Ctrl+Z) or Source Control > Discard Changes on the specific CSS file. Discarding removes uncommitted edits in that file, so check first.
- The public repository is for obtaining the starter application. Keep practice edits in your own local copy and do not push them to the shared main branch. Your teacher will explain how to submit your work separately.

6. Optional: Update your copy before another session

If the teacher updates the public repository, first save or back up your local changes. Pulling can conflict with edited files. In the project terminal, run:

```
git pull origin main
```

If Git reports a conflict, do not force-push or delete files. Ask the teacher or Copilot to explain how to resolve it safely.